Range: Last 30 Days

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Municipal Road Infrastructure Condition Report

Reporting window: Last 30 Days

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1. Executive Summary

This report provides an analysis of municipal road infrastructure conditions for the period of the Last 30 Days. A total of 166 road infrastructure events were detected and analyzed with an average confidence level of 72.8%. Potholes constitute the majority of detected issues, accounting for 111 events, while 55 speed breakers were identified. Severity distribution indicates 34 high-severity events, 63 medium-severity events, and 69 low-severity events. Key areas requiring immediate attention due to high event density and risk scores include NGOs Colony, Kamalanagara, Bengaluru; Varthur Road, Marathahalli, Bengaluru; and Service Road, HSR Layout, Bengaluru. The system currently tracks 9 active tickets for unique pothole instances, with 6 tickets resolved within the reporting period.

2. Road Health Overview

During the reporting window, a comprehensive assessment identified 166 distinct road infrastructure events across 41 unique locations. The event density was calculated at 4 events per location. All 166 detected events were successfully located.

Event Type Distribution:

- Potholes: 111 events
- Speed Breakers: 55 events

Severity Distribution:

- High Severity: 34 events
- Medium Severity: 63 events
- Low Severity: 69 events

The average confidence level for all detected events was 72.8%.

3. Detection Statistics

The AI Road Analytics Platform detected 166 road infrastructure events with an average confidence score of 72.8% over the Last 30 Days. These events were distributed across 41 distinct geographical locations, resulting in an event density of 4 events per location. All 166 detected events were successfully geolocated, with zero unlocated events.

Regarding maintenance operations, 15 unique pothole instances have been identified for ticketing. Currently, 9 active tickets are open for these issues, while 6 tickets have been resolved within the reporting period.

4. Most Critical Roads

The following roads exhibited the highest event counts during the reporting period, indicating significant infrastructure degradation or notable features:

- NGOs Colony, Kamalanagara, Bengaluru (12.995078, 77.526238): 28 events (15 potholes, 13 speed breakers). Dominant severity: Low. Risk score: 55.
- Service Road, HSR Layout, Bengaluru (12.917738, 77.623773): 14 events (14 potholes, 0 speed breakers). Dominant severity: Low. Risk score: 26.
- Varthur Road, Marathahalli, Bengaluru (12.956792, 77.701112): 12 events (11 potholes, 1 speed breaker).

Dominant severity: Medium. Risk score: 28.

- Justice Bheemiah Layout, KR Puram, Bengaluru (13.007027, 77.695966): 11 events (10 potholes, 1 speed breaker). Dominant severity: Low. Risk score: 19.
- Amarjyothi Nagara, Hebbal, Bengaluru (13.035798, 77.597001): 10 events (9 potholes, 1 speed breaker). Dominant severity: Medium. Risk score: 20.

5. High Severity Areas

Areas identified with the highest risk scores, indicating a combination of event frequency and severity, are presented below:

- NGOs Colony, Kamalanagara, Bengaluru (12.995078, 77.526238): Risk Score 55. Total 28 events (15 potholes, 13 speed breakers). Dominant severity: Low.
- Varthur Road, Marathahalli, Bengaluru (12.956792, 77.701112): Risk Score 28. Total 12 events (11 potholes, 1 speed breaker). Dominant severity: Medium.
- Service Road, HSR Layout, Bengaluru (12.917738, 77.623773): Risk Score 26. Total 14 events (14 potholes, 0 speed breakers). Dominant severity: Low.
- Amarjyothi Nagara, Hebbal, Bengaluru (13.035798, 77.597001): Risk Score 20. Total 10 events (9 potholes, 1 speed breaker). Dominant severity: Medium.
- Justice Bheemiah Layout, KR Puram, Bengaluru (13.007027, 77.695966): Risk Score 19. Total 11 events (10 potholes, 1 speed breaker). Dominant severity: Low.

6. Traffic Safety Analysis

The presence of 111 potholes and 55 speed breakers across the municipal road network contributes to potential traffic safety hazards. Of particular concern are the 34 events classified as high severity, which pose immediate risks to vehicle integrity and occupant safety.

Hotspot areas requiring focused safety interventions include:

- NGOs Colony, Kamalanagara, Bengaluru (12.995078, 77.526238): 28 events, including 15 potholes.
- Varthur Road, Marathahalli, Bengaluru (12.956792, 77.701112): 12 events, including 11 potholes.
- Service Road, HSR Layout, Bengaluru (12.917738, 77.623773): 14 events, all potholes.
- Amarjyothi Nagara, Hebbal, Bengaluru (13.035798, 77.597001): 10 events, including 9 potholes.
- Justice Bheemiah Layout, KR Puram, Bengaluru (13.007027, 77.695966): 11 events, including 10 potholes.

These locations, characterized by high event counts and risk scores, warrant priority assessment for their impact on traffic flow and accident potential.

7. Maintenance Priority Ranking

Based on the calculated risk scores and dominant severity, the following areas are ranked for maintenance priority:

1. NGOs Colony, Kamalanagara, Bengaluru (12.995078, 77.526238): Risk Score 55, Dominant Severity: Low.
2. Varthur Road, Marathahalli, Bengaluru (12.956792, 77.701112): Risk Score 28, Dominant Severity: Medium.
3. Service Road, HSR Layout, Bengaluru (12.917738, 77.623773): Risk Score 26, Dominant Severity: Low.
4. Amarjyothi Nagara, Hebbal, Bengaluru (13.035798, 77.597001): Risk Score 20, Dominant Severity: Medium.
5. Justice Bheemiah Layout, KR Puram, Bengaluru (13.007027, 77.695966): Risk Score 19, Dominant Severity: Low.

This ranking prioritizes areas with the highest cumulative risk, considering both event frequency and severity.

8. Municipal Recommendations

1. Target High Severity Events: Prioritize inspection and repair of the 34 identified high-severity events to mitigate immediate safety risks.
2. Address Pothole Accumulation: Implement a focused repair program for the 111 detected potholes, particularly in areas with high concentrations.
3. Monitor Active Tickets: Ensure timely resolution of the 9 active tickets to reduce backlog and improve public satisfaction.
4. Strategic Maintenance Planning: Utilize the identified most critical roads and high-risk areas for strategic planning of future maintenance cycles and resource allocation.
5. Data-Driven Interventions: Leverage the consistent data collection and analysis to inform proactive maintenance strategies rather than reactive repairs.

9. Predicted Impact

Failure to address the identified road infrastructure deficiencies, particularly the 34 high-severity events and the 111 potholes, is predicted to result in:

- Increased operational costs for vehicle owners due to damage.
- Elevated risk of traffic accidents and associated injuries.
- Further degradation of road surfaces, leading to more extensive and costly repairs in the future.
- Negative public perception regarding municipal infrastructure maintenance.
- Potential for increased traffic congestion due to evasive maneuvers around defects.

10. Suggested Immediate Actions

1. Emergency Response: Dispatch maintenance crews to address all 34 high-severity events within 72 hours.
2. Pothole Repair Blitz: Initiate a targeted repair program for potholes in NGOs Colony, Kamalanagara, Bengaluru; Service Road, HSR Layout, Bengaluru; and Varthur Road, Marathahalli, Bengaluru, given their high event counts and risk scores.
3. Ticket Resolution Acceleration: Expedite the resolution process for the 9 active tickets, aiming for completion within the next 14 days.
4. Field Verification: Conduct on-site verification of conditions in the top 5 highest risk areas to confirm dominant severity and inform detailed repair plans.

11. Conclusion

The analysis of road infrastructure conditions over the Last 30 Days reveals 166 events, with potholes being the predominant issue. A significant number of high and medium severity events, coupled with identified high-risk areas, necessitate prompt and strategic intervention. By prioritizing repairs in critical locations and addressing active maintenance tickets, the municipality can enhance road safety, extend infrastructure lifespan, and improve overall public service delivery. Continued utilization of AI-driven analytics will facilitate proactive maintenance and optimized resource allocation for sustainable road network management.